

Oracle Comparator Circuits versus Oracle Catalytic Logspace

1 Definitions

Definition (Comparator circuits and CC). A comparator gate is the two-input, two-output Boolean function

$$C : \{0, 1\}^2 \longrightarrow \{0, 1\}^2, \quad C(p, q) = (p \wedge q, p \vee q). \quad (1)$$

A comparator circuit on m wires is specified by a finite sequence

$$(i_1, j_1), \dots, (i_t, j_t), \quad (2)$$

where gate (i_k, j_k) applies C to the current contents of wires i_k and j_k . Each gate replaces the two values by their minimum and maximum and does not copy either input. Thus the model has fan-out one.

An annotated comparator circuit designates one output wire and labels each input wire by a constant 0 or 1, an input variable x_i , or a negated input variable $\neg x_i$. A family $\{C_n\}_{n \geq 1}$ is AC^0 -uniform when the description of C_n is generated by a function in uniform FAC^0 . The class CC consists of the relations computed by AC^0 -uniform families of polynomial-size annotated comparator circuits.

Definition (Catalytic logspace and CL). A catalytic Turing machine is a deterministic Turing machine equipped with a read-only input tape, an ordinary read-write work tape, and an additional read-write auxiliary tape. For an input x and an arbitrary string a initially written on the auxiliary tape, let $M(x, a)$ denote the resulting computation. The machine decides a language L if, for every x and a ,

1. $M(x, a)$ halts with the auxiliary tape restored exactly to a ; and
2. $M(x, a)$ accepts if and only if $x \in L$.

For functions $S, S_a : \mathbb{N} \rightarrow \mathbb{N}$, define

$$\text{CSPACE}(S(n), S_a(n)) \quad (3)$$

as the class of languages decided by catalytic machines using $O(S(n))$ cells of ordinary work tape and $O(S_a(n))$ cells of auxiliary tape on inputs of length n . Write

$$\text{CSPACE}(S(n)) := \text{CSPACE}(S(n), 2^{O(S(n))}), \quad \text{CL} := \text{CSPACE}(\log n). \quad (4)$$

Definition (Native functional oracle for comparator circuits). A comparator-circuit oracle is a length-preserving function

$$\alpha : \{0, 1\}^* \longrightarrow \{0, 1\}^*, \quad |\alpha(y)| = |y|. \quad (5)$$

For each n , let α_n be the restriction of α to $\{0, 1\}^n$. An α_n -gate in a comparator circuit selects n distinct wires carrying $y \in \{0, 1\}^n$ and sends the n bits of $\alpha_n(y)$ to n distinct outgoing wires. Each output is connected to at most one later gate input, so the limited-fan-out condition is preserved.

For an input X of length n , a relation $R(X, \alpha)$ belongs to $\text{CC}(\alpha)$ when it is computed by a polynomial-size uniform family of annotated comparator circuits whose allowed gates are comparator gates, negation gates, and α_n -gates.

Definition (Weak oracle for catalytic logspace). Let $O : \{0, 1\}^* \rightarrow \{0, 1\}$ be a Boolean oracle. A CL^O machine has a one-bit read-only oracle-output register, initially 0, but no query tape. An oracle call designates a subinterval T of data already present on the input or catalytic tape. If y is the current content of T , the call performs

$$\text{oracle-output} \leftarrow O(y) \quad (6)$$

without modifying T . The two endpoints of T are encoded using $O(\log n)$ bits of the machine's existing storage. For every input and every initial catalyst, the machine must return the correct answer and restore the catalyst exactly.

Definition (Exact convention for CC^O with a Boolean oracle). To compare comparator circuits with CL^O relative to one and the same Boolean oracle, fix

$$O : \{0, 1\}^* \longrightarrow \{0, 1\} \quad (7)$$

and associate with it the length-preserving function $\widehat{O}$ given by

$$\widehat{O}(\varepsilon) = \varepsilon, \quad \widehat{O}(by) = (b \oplus O(y))y \quad (b \in \{0, 1\}, y \in \{0, 1\}^*). \quad (8)$$

Thus $|\widehat{O}(z)| = |z|$. On $r = m + 1$ wires, a $\widehat{O}_r$ -gate acts as

$$(b, y_1, \dots, y_m) \mapsto (b \oplus O(y_1 \cdots y_m), y_1, \dots, y_m). \quad (9)$$

It exposes $O(y)$ when its answer wire is initialized to $b = 0$, preserves the query bits, has equally many inputs and outputs, and creates no fan-out.

For this paper, define

$$\boxed{\text{CC}^O := \text{CC}(\widehat{O})}. \quad (10)$$

Equivalently, a language A belongs to CC^O if there is an AC^0 -uniform polynomial-size family of annotated comparator circuits, independent of the values of O , using comparator gates, negation gates, and the corresponding $\widehat{O}_n$ -gates, whose designated output is 1 exactly on the strings in L . This reversible Boolean embedding is the comparison convention adopted here.

2 Best Known Result

The unrelativized containments are

$$\text{NL} \subseteq \text{CC} \subseteq \text{P}, \quad \text{NL} \subseteq \text{CL} \subseteq \text{ZPP}. \quad (11)$$

In the native functional-oracle model for comparator circuits, there are suitable oracle settings α and β , not necessarily the same, for which

$$\text{CC}(\alpha) \not\subseteq \text{NC}(\alpha), \quad \text{NC}^3(\beta) \not\subseteq \text{CC}(\beta). \quad (12)$$

These separations remain valid when the functional oracles are restricted to be strictly 1-Lipschitz. The relativization of NC used in this model controls the nesting depth of oracle gates and preserves containments such as

$$\text{NC}^1(\alpha) \subseteq \text{L}(\alpha). \quad (13)$$

These results do not separate the common Boolean-oracle classes appearing in the problem below, because an arbitrary length-preserving functional oracle need not arise from a Boolean oracle.

For oracle catalytic logspace, there are oracle worlds in which the class collapses to ordinary logspace and others in which, for a suitable O ,

$$\text{CL}^O = \text{PSPACE}^O. \quad (14)$$

The strongest result stated in the source is

$$\exists O \quad \text{CL}^O = \text{EXP}^O, \quad (15)$$

even under the weak no-query-tape convention above.

3 Problem Formalization

Prove that there are oracles O_1 and O_2 such that

$$\text{CC}^{O_1} \not\subseteq \text{CL}^{O_1} \quad \text{and} \quad \text{CL}^{O_2} \not\subseteq \text{CC}^{O_2}. \quad (16)$$

Then prove the stronger version: there exists a single oracle O such that

$$\text{CC}^O \not\subseteq \text{CL}^O \quad \text{and} \quad \text{CL}^O \not\subseteq \text{CC}^O. \quad (17)$$